BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Hexadecimal

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. Previous lesson: what is 1 + 1 in binary?

2. How many bits make one byte?

3. Convert 10110000 to denary. Show place values.

4. Why is 255 the highest unsigned eight-bit value?

Today's objectives

Convert between binary, denary and hex; split a byte into two nibbles.

Key vocabulary - Read, cover, recall

hexadecimal
Base 16: digits 0-9 and A-F

nibble
Four binary bits

digit
One symbol in a number

conversion
Writing a value in another base

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Four bits make one hex digit

Hexadecimal uses 0-9 and A-F. A=10, B=11, C=12, D=13, E=14, F=15.

A byte becomes two nibbles. Each nibble uses the 8, 4, 2, 1 values.

Nibble 1	Nibble 2
1111 = F	**1110 = E**

11111110 becomes FE. Hex is shorter to read; bits are unchanged.

Revision-guide example - RGB colour #00FF00

RED	GREEN	BLUE
00	FF	00

I Do - Convert binary to hex

11000111 = 1100 | 0111. Use 8,4,2,1 within each nibble.

1100 = 8+4 = 12 = C; 0111 = 4+2+1 = 7; Answer = C7.

I Do - Hex to denary

A3 = (10 x 16) + 3 = 163. The left digit counts sixteens.

We Do - Guided practice

Convert 01101101 to hex: 0110 | 1101 = ____ | ____ = _____.

Left nibble = _____ Right nibble = _____

Complete hexadecimal = _____

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

Start on paper. Do not look at model answers until page 4.

```
value = 6 * 16 + 2
print(value)
```

Predict: What does print(11 * 16 + 0) output?

Write code to display the denary value of hex A3.

Change the starter to display the denary value of C7.

You Do - Independent work

Convert 10110000 from binary to hex. Show nibbles.

Convert hexadecimal A3 to denary with working.

Convert denary 98 to hexadecimal. Show two nibbles.

Thinking: A pupil says hex 1E = 15. Explain why that is wrong.

Exit Ticket - True or False?

Answer before you turn to the green-pen page.

1. One hex digit represents four bits. ☐ T ☐ F
2. E represents 15 in hexadecimal. ☐ T ☐ F
3. Hex reduces the number of stored bits. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick correct answers. For a mistake, use the model answer then copy the FULL question and write a fresh answer below.

Do Now - Answers

Q1 10 in binary. ☐

Q2 8 bits. ☐

Q3 128 + 32 + 16 = 176. ☐

Q4 256 patterns numbered 0 through 255. ☐

We Do - Guided practice

0110 | 1101 = 6 | D = 6D

Key vocabulary

hexadecimal ☐
Base 16: digits 0-9 and A-F

nibble ☐
Four binary bits

digit ☐
One symbol in a number

conversion ☐
Writing a value in another base

Code First - Answers

Predict ☐
176

Write ☐
print(10 * 16 + 3) gives 163

Improve ☐
value = 12 * 16 + 7; print(value) gives 199

Exit Ticket - Explanations

1. True

2. False: E = 14; F = 15

3. False: only notation is shorter

You Do - Self-mark

Q1	1011 | 0000 = B0	☐
Q2	10 x 16 + 3 = 163	☐
Q3	0110 | 0010 = 62	☐
Thinking	The left digit counts 16s: 16 + 14 = 30	☐

Try the question again

1. Copy the FULL question you got wrong (or the hardest):

2. In GREEN PEN, write a NEW answer. Show working: